AquaGuard AI — Concept Note & Lean Canvas

AI-Powered Water Quality Screening & Regulatory Decision Support System | Dual-Layer GBDT & BIS IS 10500 Engine

PROJECT CONCEPT NOTE
UN SDG 6 ALIGNED

1. PROBLEM

- **Lab Latency:** Statutory water testing turnaround takes 3–7 days, delaying emergency contamination alerts.
- **Testing Expense:** Running comprehensive assays for every wellhead/tank is economically unfeasible.
- **Field Triage Vacuum:** Field officers record probe telemetry (pH, turbidity, TDS) but lack decision support.
- **Black-Box Trust Deficit:** Generic ML classifiers provide opaque scores without legal standard mapping.

4. SOLUTION

- **Dual-Layer Screening:** Calibrated GBDT ML + BIS IS 10500 & WHO compliance mapping.
- **Explainable AI (SHAP):** Real-time parameter-level attribution graphs.
- **High-Throughput Batch:** Validates & scores 500 field samples via CSV drag-and-drop.

8. KEY METRICS

- **Turnaround:** < 200 ms per sample.
- **Model Performance:** ROC-AUC 0.924, Accuracy 84.0%, Brier Score 0.095.
- **Lab Prioritization:** 60% reduction in unnecessary statutory testing dispatches.

3. UNIQUE VALUE PROPOSITION

- **Rapid Field Triage in Seconds:** Converts 9 physicochemical probe readings into calibrated risk tiers.
- **Explainable Over Black-Box:** Transparent SHAP feature bars prove *why* a sample was flagged.
- **Responsible AI by Design:** Strictly non-diagnostic; refuses false drinkability claims and prioritizes lab referral.
- **Zero LLM Hallucinations:** Deterministic geochemical ML pipeline.

9. UNFAIR ADVANTAGE

- **Dual-Engine Architecture:** Simultaneously checks statistical geochemical vectors & formal BIS legal limits.
- **Zero-Config Deployment:** Single-terminal unified runner, offline SQLite persistence, and local mock fallback.

5. CHANNELS

- **Government Water Boards:** Public health surveillance (Jal Jeevan Mission / CPCB).
- **Municipal Utilities:** Urban water supply & treatment plant monitoring.
- **Open APIs:** Direct RESTful ingestion for IoT water sensor gateways.

2. CUSTOMER SEGMENTS

- **Municipal Water Boards:** Urban pipeline grid and water treatment operators.
- **Rural Water Missions:** Field personnel monitoring decentralized borewells.
- **Industrial Effluent Monitors:** Environmental audit officers checking factory discharge.
- **Watershed NGOs:** Citizen science teams conducting community catchment surveys.

7. COST STRUCTURE

- **Compute & Infrastructure:** FastAPI backend, SQLite database, and static CDN hosting.
- **ML Pipeline & Calibration:** Continuous retraining against empirical regional geochemical surveys.
- **Regulatory Maintenance:** Ongoing updates to Bureau of Indian Standards (BIS IS 10500) benchmark tables.
- **Quality Assurance:** Automated unit tests, security audits, and sensor variance checks.

6. REVENUE STREAMS & IMPACT MODEL

- **SaaS Subscriptions for Utilities:** Tiered municipal access for watershed surveillance dashboards & batch features.
- **Telemetry Gateway API Licensing:** Usage-based API tiers for hardware sensor manufacturers & IoT data loggers.
- **Accredited Lab Referral Integration:** Digital dispatch partnerships with certified testing facilities.
- **UN SDG 6 Social Impact:** Prevents waterborne health hazards by accelerating early contamination detection.

Educational & Decision-Support Notice: *AquaGuard AI is an automated screening triage tool and does not certify water as safe or potable. It is designed to assist environmental surveillance and prioritize samples for statutory, accredited laboratory testing.*